

Double/Debiased Machine Learning Extension of Nunn (2008)

The Long-Term Effects of Africa’s Slave Trades: A Causal ML Replication

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1. Research Question

Can machine learning strengthen a causal replication of Nunn (2008)?

Original question: Did Africa’s slave trades have persistent negative effects on long-run economic development?

ML extension: Apply DML-IV while retaining Nunn’s original IV identification strategy and allowing for flexible adjustment for colonial and geographic controls.

Core Contribution

Machine learning is used for **nuisance estimation**, not for replacing the IV design. Causal identification rely on Nunn’s original instrumental variables and causal framework.

2. Original Paper and Replication

Paper: Nunn (2008), “The Long-Term Effects of Africa’s Slave Trades.”

Outcome: log real GDP per capita in 2000.

Treatment: log slave exports normalized by land area.

Instruments: distances to slave demand market: Atlantic, Indian Ocean, Saharan, and Red Sea.

Sample: 52 African countries.

Controls: 39 variables including geographical and historical information.

Method	Estimate	Std. Error
OLS	−0.076***	(0.029)
2SLS	−0.286*	(0.153)

Replication note: The main OLS and IV specifications are replicated exactly. However, after restricting the sample in the IV analysis, the replicated coefficient differs in magnitude from that reported in the original paper.

3. Why DML-IV?

- DML combines machine learning with causal inference by flexibly estimating **nuisance functions** while preserving valid estimation of the causal effect.
- Nunn’s original causal design uses historical distance instruments to identify the effect of slave exports on income.
- DML-IV retains this IV structure but allows the outcome, treatment, and instrument equations to adjust flexibly for colonial and geographic controls.
- Cross-fitting reduces overfitting bias and improves the reliability of causal estimation.

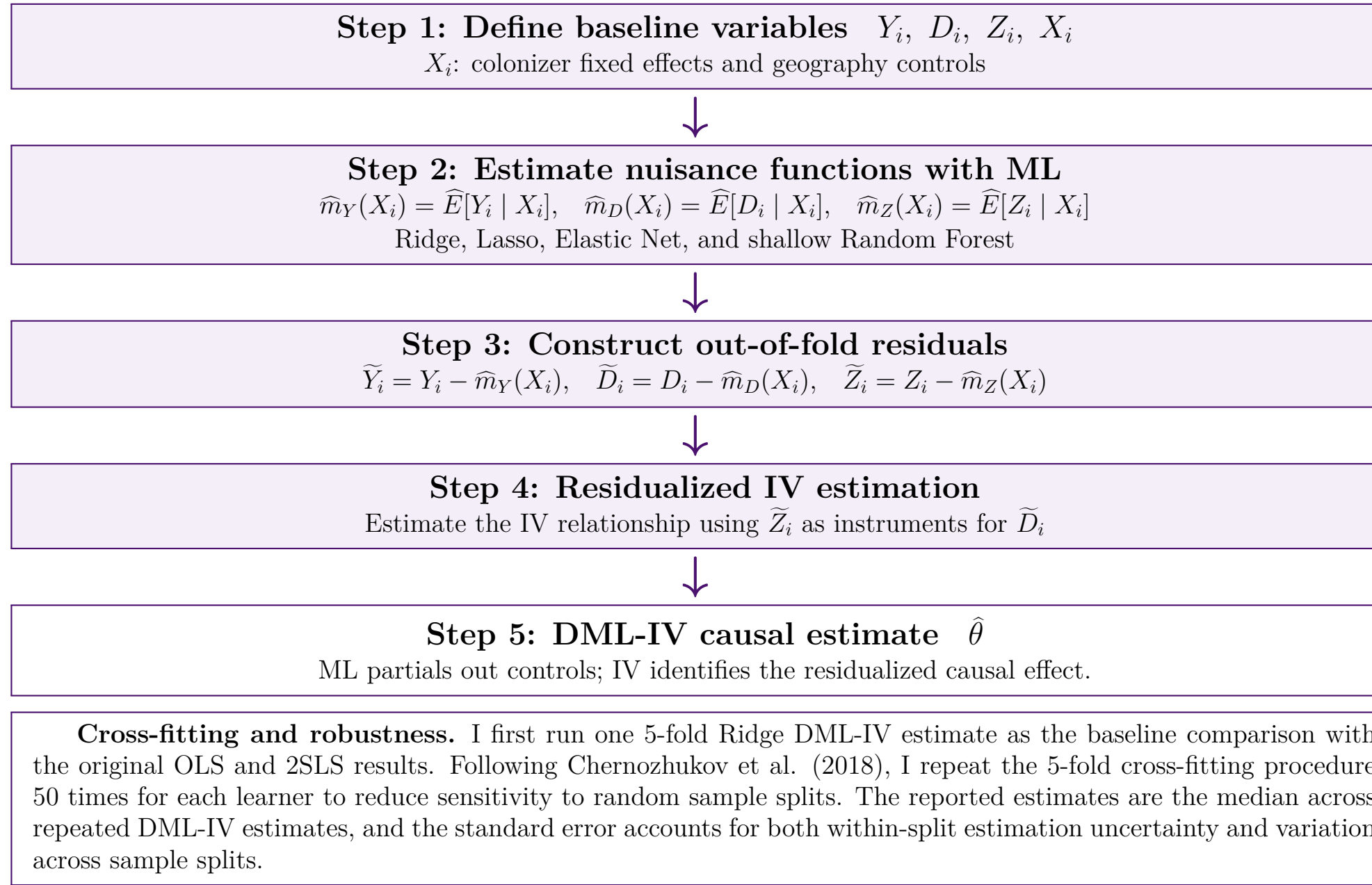
4. Machine Learning Learners

Learner	Role in DML-IV
Ridge	L_2 shrinkage. Conservative baseline that keeps all controls but shrinks coefficients.
Lasso	L_1 penalty. Allows sparse nuisance prediction and variable selection.
Elastic Net	Combines L_1 and L_2 penalties. Hybrid between shrinkage and variable selection.
Shallow RF	Nonlinear learner. Uses 100 trees, max depth 2, and min leaf size 5 to avoid overfitting.

Important Design

All learners estimate $E[Y|X]$, $E[D|X]$, and $E[Z|X]$ only. They do **not** replace Nunn’s distance instruments. The final causal estimate still comes from an IV regression.

5. DML-IV Workflow



6. Baseline DML-IV Result

	OLS	2SLS	DML-IV Ridge
ln(exports/area)	−0.076***	−0.286*	−0.144*
Std. error	(0.029)	(0.153)	(0.085)
First-stage F	–	0.811	1.316
Observations	52	52	52

Interpretation: why ridge: small sample; shrinks the magnitude coefficients; remain all variables, conservative results. The Ridge-based DML-IV estimate remains negative and significant at 1% level and weak instrument variables.

7. Learner Robustness: 50 Repeated Splits

Learner	Median $\hat{\theta}$	Adj. SE	Adj. p	Median F	Share < 0
Ridge	−0.240**	0.106	0.023	1.352	1.00
Lasso	−0.237**	0.097	0.015	1.721	1.00
Elastic Net	−0.260**	0.113	0.021	1.333	1.00
Shallow RF	−0.249***	0.088	0.005	2.139	1.00

Notes: Results summarize 50 repeated five-fold DML-IV sample splits. Median F is the median first-stage F -statistic across splits. Share < 0 is the proportion of repeated estimates that are negative. Adjusted standard errors account for variation across sample splits. * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Restricted-sample check (N=42): Using the same restricted sample as Table IV column (4), all four DML-IV learners remain negative across 50 repeated splits. Median estimates: Ridge −0.200, Lasso −0.214, Elastic Net −0.210, Shallow RF −0.246. For all learners, Share $< 0 = 1.00$.

8. Limitations and Takeaway

Limitations

- The small sample size (N=52) constrains the use of more complex machine-learning methods.
- Weak instruments remain a concern.DML improves nuisance adjustment, not the validity or strength of the instruments.

Main Takeaway

Machine learning reduces the influence of noisy controls and sensitivity to a single specification. Cross-fitting reduces overfitting. The IV estimates remain stable across learners.

References:

Chernozhukov et al. (2018), “Double/debiased machine learning for treatment and structural parameters.”
Nunn (2008), “The Long-Term Effects of Africa’s Slave Trades.”